

HeveMind Model Monitoring Report

Industrial monitoring of sensor distributions, prediction outputs, decision patterns, confidence, uncertainty and data quality. This module does not change the deployed model.

OVERALL HEALTH	HEALTH BAND	RECOMMENDATION	CRITICAL ALERTS	MODEL MODIFIED
87.89/100	Watch	Immediate Retraining	6	No

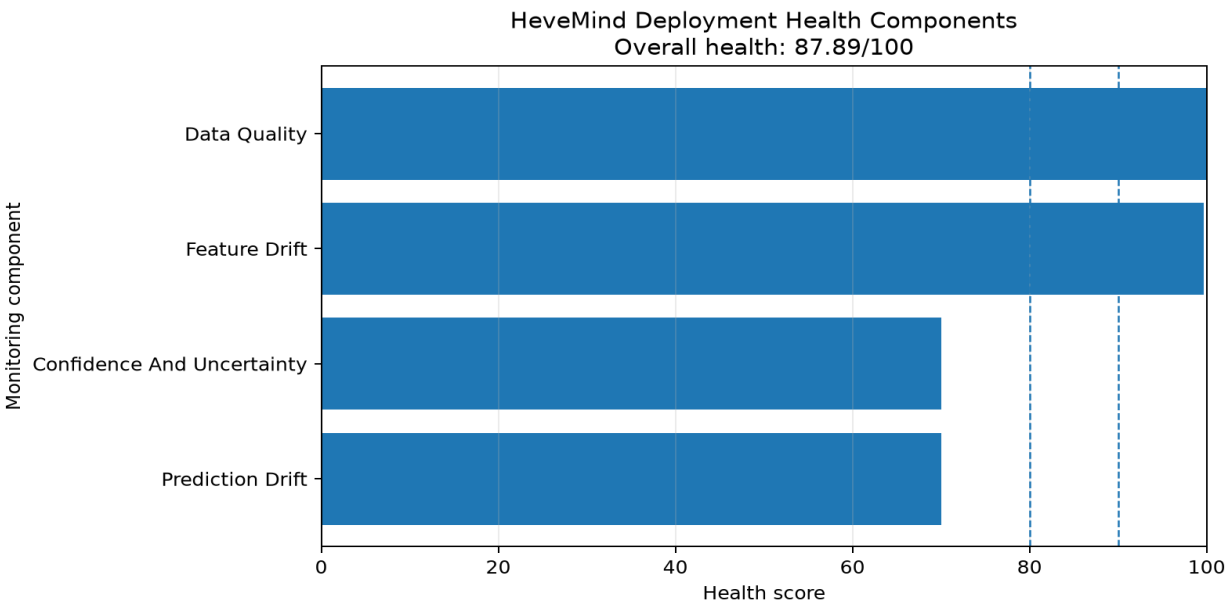
Executive Assessment

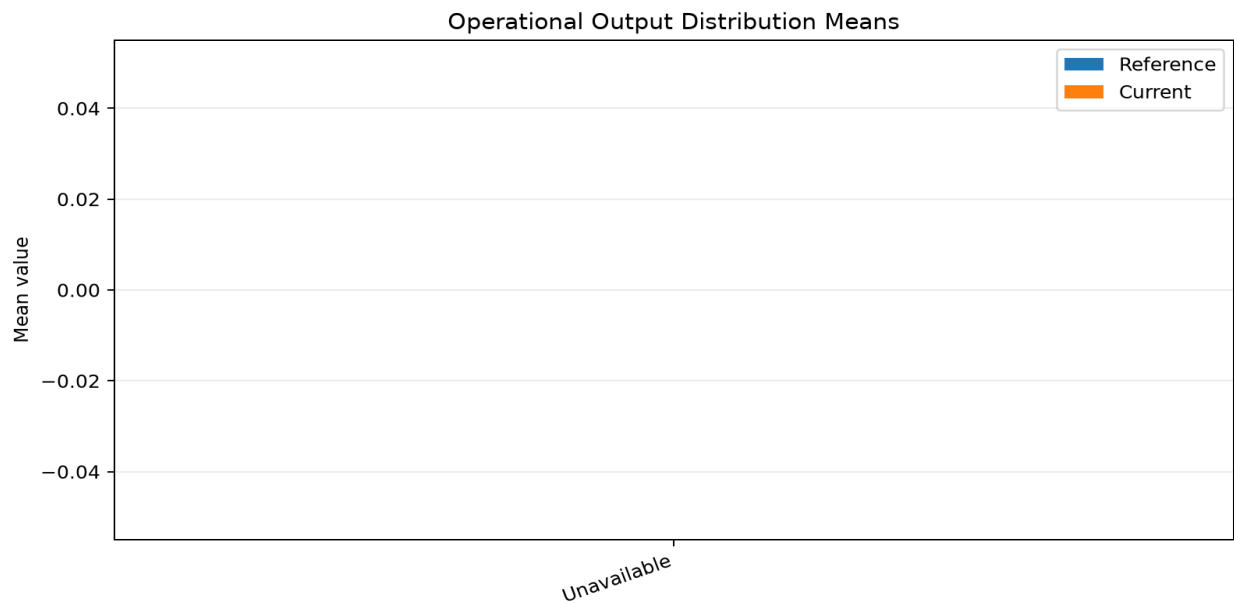
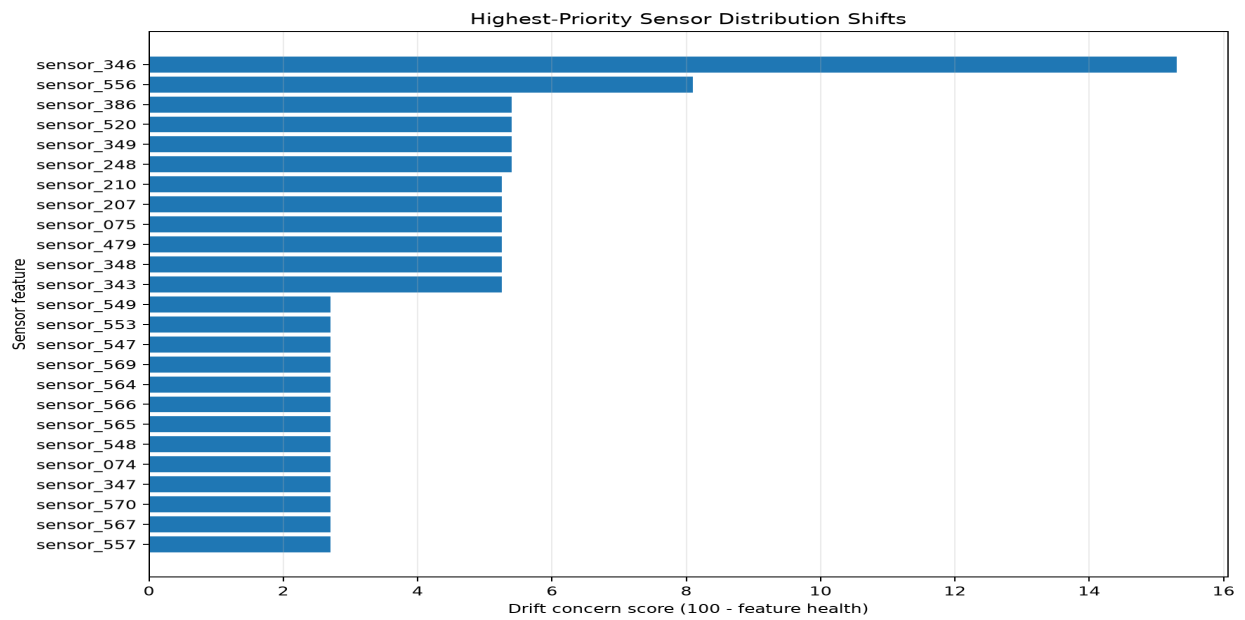
Deployment health is Watch at 87.89/100. The primary risk driver is Prediction Drift (70.00/100, Warning). Recommendation: Immediate Retraining.

Production status: Escalated engineering and model-owner review required. Reason: Multiple critical monitoring signals indicate that the deployed model may no longer represent the current process.

Health Components

Component	Score	Weight	Severity	Explanation
Data Quality	100.00	0.25	Normal	Covers missingness, duplication, constant sensors, invalid probability-like values and schema mismatch.
Feature Drift	99.70	0.35	Critical	34 of 450 monitored features show Watch-or-higher drift.
Prediction Drift	70.00	0.25	Warning	Monitors failure probability and other available operational output distributions.
Confidence And Uncertainty	70.00	0.15	Warning	Measures drift in prediction confidence, uncertainty and available input-data confidence.





Priority Monitoring Alerts

Severity	Category	Resource	Alert	Recommended action
Critical	Feature Drift	sensor_075	sensor_075 is classified as Critical for distribution drift (PSI=0.0000, KS=0.0008, standardised mean shift=0.0283, missing-rate difference=0.0032).	Escalate process and sensor review, compare with maintenance and recipe changes, and assess retraining immediately.
Critical	Feature Drift	sensor_207	sensor_207 is classified as Critical for distribution drift (PSI=0.0000, KS=0.0008, standardised mean shift=0.0283, missing-rate difference=0.0032).	Escalate process and sensor review, compare with maintenance and recipe changes, and assess retraining immediately.
Critical	Feature Drift	sensor_210	sensor_210 is classified as Critical for distribution drift (PSI=0.0000, KS=0.0008, standardised mean shift=0.0283, missing-rate difference=0.0032).	Escalate process and sensor review, compare with maintenance and recipe changes, and assess retraining immediately.
Critical	Feature Drift	sensor_343	sensor_343 is classified as Critical for distribution drift (PSI=0.0000, KS=0.0008, standardised mean shift=0.0283, missing-rate difference=0.0032).	Escalate process and sensor review, compare with maintenance and recipe changes, and assess retraining immediately.
Critical	Feature Drift	sensor_348	sensor_348 is classified as Critical for distribution drift (PSI=0.0000, KS=0.0008, standardised mean shift=0.0283, missing-rate difference=0.0032).	Escalate process and sensor review, compare with maintenance and recipe changes, and assess retraining immediately.
Critical	Feature Drift	sensor_479	sensor_479 is classified as Critical for distribution drift (PSI=0.0000, KS=0.0008, standardised mean shift=0.0283, missing-rate difference=0.0032).	Escalate process and sensor review, compare with maintenance and recipe changes, and assess retraining immediately.
Warning	Feature Drift	sensor_346	sensor_346 is classified as Warning for distribution drift (PSI=0.2317, KS=0.0648, standardised mean shift=0.1319, missing-rate difference=0.0792).	Increase monitoring frequency, investigate the affected evidence and prepare a controlled retraining assessment.
Warning	Prediction Drift	combined_uncertainty	Combined Uncertainty is classified as Warning; reference mean=None, current mean=None.	Increase monitoring frequency, investigate the affected evidence and prepare a controlled retraining assessment.
Warning	Prediction Drift	data_confidence	Data Confidence is classified as Warning; reference mean=None, current mean=None.	Increase monitoring frequency, investigate the affected evidence and prepare a controlled retraining assessment.
Warning	Prediction Drift	failure_probability	Failure Probability is classified as Warning; reference mean=None, current mean=None.	Increase monitoring frequency, investigate the affected evidence and prepare a controlled retraining assessment.
Warning	Prediction Drift	prediction_confidence	Prediction Confidence is classified as Warning; reference mean=None, current mean=None.	Increase monitoring frequency, investigate the affected evidence and prepare a controlled retraining assessment.
Watch	Feature Drift	sensor_073	sensor_073 is classified as Watch for distribution drift (PSI=0.0582, KS=0.0929, standardised mean shift=0.1399, missing-rate difference=0.0792).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_074	sensor_074 is classified as Watch for distribution drift (PSI=0.0382, KS=0.0516, standardised mean shift=0.1109, missing-rate difference=0.0792).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_248	sensor_248 is classified as Watch for distribution drift (PSI=0.1154, KS=0.0708, standardised mean shift=0.0150, missing-rate difference=0.0148).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_347	sensor_347 is classified as Watch for distribution drift (PSI=0.0400, KS=0.0563, standardised mean shift=0.1336, missing-rate difference=0.0792).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_349	sensor_349 is classified as Watch for distribution drift (PSI=0.1033, KS=0.0909, standardised mean shift=0.1276, missing-rate difference=0.0032).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_386	sensor_386 is classified as Watch for distribution drift (PSI=0.1016, KS=0.0503, standardised mean shift=0.0046, missing-rate difference=0.0148).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_520	sensor_520 is classified as Watch for distribution drift (PSI=0.1027, KS=0.0685, standardised mean shift=0.0242, missing-rate difference=0.0148).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.

Severity	Category	Resource	Alert	Recommended action
Watch	Feature Drift	sensor_547	sensor_547 is classified as Watch for distribution drift (PSI=0.0097, KS=0.0420, standardised mean shift=0.0876, missing-rate difference=0.0315).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_548	sensor_548 is classified as Watch for distribution drift (PSI=0.0306, KS=0.0719, standardised mean shift=0.1121, missing-rate difference=0.0315).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_549	sensor_549 is classified as Watch for distribution drift (PSI=0.0457, KS=0.0676, standardised mean shift=0.0692, missing-rate difference=0.0315).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_550	sensor_550 is classified as Watch for distribution drift (PSI=0.0458, KS=0.0576, standardised mean shift=0.0606, missing-rate difference=0.0315).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_551	sensor_551 is classified as Watch for distribution drift (PSI=0.0502, KS=0.0674, standardised mean shift=0.0161, missing-rate difference=0.0315).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_552	sensor_552 is classified as Watch for distribution drift (PSI=0.0486, KS=0.0874, standardised mean shift=0.0807, missing-rate difference=0.0315).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_553	sensor_553 is classified as Watch for distribution drift (PSI=0.0422, KS=0.0494, standardised mean shift=0.0596, missing-rate difference=0.0315).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_554	sensor_554 is classified as Watch for distribution drift (PSI=0.0477, KS=0.0753, standardised mean shift=0.0202, missing-rate difference=0.0315).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_555	sensor_555 is classified as Watch for distribution drift (PSI=0.0664, KS=0.0882, standardised mean shift=0.0790, missing-rate difference=0.0315).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_556	sensor_556 is classified as Watch for distribution drift (PSI=0.1004, KS=0.0450, standardised mean shift=0.0118, missing-rate difference=0.0315).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_557	sensor_557 is classified as Watch for distribution drift (PSI=0.0411, KS=0.0714, standardised mean shift=0.0184, missing-rate difference=0.0315).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.
Watch	Feature Drift	sensor_558	sensor_558 is classified as Watch for distribution drift (PSI=0.0512, KS=0.0992, standardised mean shift=0.0764, missing-rate difference=0.0315).	Continue deployment with enhanced monitoring and document the observed change for trend analysis.

Highest-Priority Feature Drift

Feature	Severity	Health	PSI	KS	Notes
sensor_075	Critical	94.75	0.0000	0.0008	Current sensor is constant. Sensor became constant in current data. Current missingness exceeds reference missingness.
sensor_207	Critical	94.75	0.0000	0.0008	Current sensor is constant. Sensor became constant in current data. Current missingness exceeds reference missingness.
sensor_210	Critical	94.75	0.0000	0.0008	Current sensor is constant. Sensor became constant in current data. Current missingness exceeds reference missingness.
sensor_343	Critical	94.75	0.0000	0.0008	Current sensor is constant. Sensor became constant in current data. Current missingness exceeds reference missingness.
sensor_348	Critical	94.75	0.0000	0.0008	Current sensor is constant. Sensor became constant in current data. Current missingness exceeds reference missingness.
sensor_479	Critical	94.75	0.0000	0.0008	Current sensor is constant. Sensor became constant in current data. Current missingness exceeds reference missingness.
sensor_346	Warning	84.70	0.2317	0.0648	Current missingness exceeds reference missingness.
sensor_556	Watch	91.90	0.1004	0.0450	Current missingness exceeds reference missingness.
sensor_248	Watch	94.60	0.1154	0.0708	Current missingness exceeds reference missingness.

Feature	Severity	Health	PSI	KS	Notes
sensor_349	Watch	94.60	0.1033	0.0909	No additional data-quality note.
sensor_520	Watch	94.60	0.1027	0.0685	Current missingness exceeds reference missingness.
sensor_386	Watch	94.60	0.1016	0.0503	Current missingness exceeds reference missingness.
sensor_555	Watch	97.30	0.0664	0.0882	Current missingness exceeds reference missingness.
sensor_568	Watch	97.30	0.0649	0.0441	No additional data-quality note.
sensor_073	Watch	97.30	0.0582	0.0929	Current missingness exceeds reference missingness.
sensor_558	Watch	97.30	0.0512	0.0992	Current missingness exceeds reference missingness.
sensor_551	Watch	97.30	0.0502	0.0674	Current missingness exceeds reference missingness.
sensor_552	Watch	97.30	0.0486	0.0874	Current missingness exceeds reference missingness.
sensor_554	Watch	97.30	0.0477	0.0753	Current missingness exceeds reference missingness.
sensor_550	Watch	97.30	0.0458	0.0576	Current missingness exceeds reference missingness.
sensor_549	Watch	97.30	0.0457	0.0676	Current missingness exceeds reference missingness.
sensor_563	Watch	97.30	0.0436	0.0602	No additional data-quality note.
sensor_553	Watch	97.30	0.0422	0.0494	Current missingness exceeds reference missingness.
sensor_557	Watch	97.30	0.0411	0.0714	Current missingness exceeds reference missingness.
sensor_570	Watch	97.30	0.0407	0.0487	No additional data-quality note.

Prediction and Decision Monitoring

Output	Severity	Reference mean	Current mean	PSI	Health
Failure Probability	Warning	Unavailable	Unavailable	Unavailable	70.00
Prediction Confidence	Warning	Unavailable	Unavailable	Unavailable	70.00
Combined Uncertainty	Warning	Unavailable	Unavailable	Unavailable	70.00
Data Confidence	Warning	Unavailable	Unavailable	Unavailable	70.00

Data Quality

Metric	Current	Reference	Severity
Current constant-feature rate	0.0133	Unavailable	Normal
Current duplicate-row rate	0.0000	Unavailable	Normal
Current total missing rate	0.0200	0.0199	Normal
Duplicate ID rate	0.0000	Unavailable	Normal
Invalid confidence rate	Unavailable	Unavailable	Normal
Invalid probability rate	Unavailable	Unavailable	Normal
Invalid uncertainty rate	Unavailable	Unavailable	Normal
Unmatched-feature rate	0.0000	Unavailable	Normal

Interpretation limitation: monitoring detects statistical changes and operational risk signals. It does not establish a physical failure mechanism and does not automatically replace or retrain the deployed model.